

Marcel Neunhoeffner

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RESEARCH INTERESTS

Statistical inference after upstream data transformations: multiple imputation, synthetic data, differential privacy, and tabular foundation models. Political methodology, election forecasting, machine learning for the social sciences, official statistics.

ACADEMIC POSITIONS

Percentages give the share of a full-time appointment. The LMU position has run concurrently with the Boston University (2021–2023) and IAB (since 2023) positions.

Postdoctoral Research Associate (80%) Since 2023
Institute for Employment Research (IAB), Statistical Methods Unit, Nuremberg

Teaching and Research Associate (20%) Since 2021
Ludwig-Maximilians-Universität München, Department of Statistics, Social Data Science and AI Lab

Lecturer, Machine Learning 2025–2026
Diplomatische Akademie Wien

Research Associate (80%, partly in Boston, partly remote) 2021–2023
Boston University, Department of Computer Science (with Adam Smith, Marco Gaboardi, and Mark Bun)

Teaching and Research Associate (doctoral position) 2016–2021
University of Mannheim, Quantitative Methods in the Social Sciences

Lecturer, Privacy Engineering 2020
University of California, Berkeley, School of Information

Visiting Ph.D. Student 2019
Simons Institute for the Theory of Computing, UC Berkeley

EDUCATION

Ph.D. Political Science (Dr. rer. soc.) 2023
University of Mannheim, Graduate School of Economic and Social Sciences
Thesis: Generative Adversarial Nets for Social Scientists
Supervisors: Prof. Thomas Gschwend, Ph.D. and Prof. Dr. Frauke Kreuter

M.A. Political Science 2016
University of Mannheim

B.A. Governance and Public Policy 2013
University of Passau

PUBLICATIONS

Peer-Reviewed Articles and Conference Proceedings

18. Neunhoeffner, M. & Drechsler, J. (2027). Method or Implementation? A Plea for More Rigor in Synthetic Data Benchmarking. *Privacy in Statistical Databases (PSD)*

- 2026), 276–291.
17. Garg, S., **Neunhoeffler, M.**, Drechsler, J., & Torra, V. (2026). Using Prior Knowledge to Improve GANs for Tabular Data Without Compromising Privacy. *Data Privacy Management (DPM 2025), held with ESORICS 2025*, 137–153.
 16. **Neunhoeffler, M.**, Seeman, J., & Drechsler, J. (2026). On the Formal Privacy Guarantees of Synthetic Data (Generated Without Formal Privacy Guarantees). *Harvard Data Science Review*, Special Issue 6.
 15. **Neunhoeffler, M.** (2026). An Introduction to Generative Adversarial Nets in R: The RGAN Package. *The R Journal*, 18(1): 5–22.
 14. Arnold, C. & **Neunhoeffler, M.** (2025). Working with Synthetic Data: The Do’s, Dangers and Don’ts. *Sage Research Methods: Research and Data Literacy*.
 13. Breznau, N., Rinke, E. M., Wuttke, A., et al. [185 authors incl. **Neunhoeffler, M.**]. (2025). The Reliability of Replications: A Study in Computational Reproductions. *Royal Society Open Science*, 12(3): 241038.
 12. Lehrer, R., Bahnsen, O., Müller, K., **Neunhoeffler, M.**, Gschwend, T., & Juhl, S. (2025). Rallying Around the Leader in Times of Crises: The Opposing Effects of Perceived Threat and Anxiety. *European Journal of Political Research*, 64(2): 697–718.
 11. Rittmann, O., **Neunhoeffler, M.**, & Gschwend, T. (2025). How to Improve the Substantive Interpretation of Regression Results when the Dependent Variable is Logged. *Political Science Research and Methods*, 13(1): 203–211.
 10. Arnold, C., Biedebach, L., Küpfer, A., & **Neunhoeffler, M.** (2024). The Role of Hyperparameters in Machine Learning Models and How to Tune Them. *Political Science Research and Methods*, 12(4): 841–848.
 9. Bun, M., Gaboardi, M., **Neunhoeffler, M.**, & Zhang, W. (2024). Continual Release of Differentially Private Synthetic Data from Longitudinal Data Collections. *Proceedings of the ACM on Management of Data (PODS)*, 2(2), Article 94.
 8. Latner, J., **Neunhoeffler, M.**, & Drechsler, J. (2024). Generating Synthetic Data is Complicated: Know Your Data and Know Your Generator. *Privacy in Statistical Databases (PSD 2024)*, 115–128.
 7. Breznau, N., Rinke, E. M., Wuttke, A., et al. [166 authors incl. **Neunhoeffler, M.**]. (2022). Observing Many Researchers Using the Same Data and Hypothesis Reveals a Hidden Universe of Uncertainty. *Proceedings of the National Academy of Sciences*, 119(44).
 6. Gschwend, T., Müller, K., Munzert, S., **Neunhoeffler, M.**, & Stoetzer, L. F. (2022). The Zweitstimme Model: A Dynamic Forecast of the 2021 German Federal Election. *PS: Political Science & Politics*, 55(1): 85–90.
 5. **Neunhoeffler, M.**, Wu, Z. S., & Dwork, C. (2021). Private Post-GAN Boosting. *Proceedings of the 9th International Conference on Learning Representations (ICLR)*.
 4. **Neunhoeffler, M.**, Gschwend, T., Munzert, S., & Stoetzer, L. F. (2020). Ein Ansatz zur Vorhersage der Erststimmenanteile bei Bundestagswahlen. *Politische Vierteljahresschrift*, 61(1): 111–130.
 3. **Neunhoeffler, M.** & Sternberg, S. (2019). How Cross-Validation Can Go Wrong and What to Do About It. *Political Analysis*, 27(1): 101–106.
 2. Stoetzer, L. F., **Neunhoeffler, M.**, Gschwend, T., Munzert, S., & Sternberg, S. (2019). Forecasting Elections in Multiparty Systems: A Bayesian Approach Combining Polls and Fundamentals. *Political Analysis*, 27(2): 255–262.
 1. Munzert, S., Stötzer, L., Gschwend, T., **Neunhoeffler, M.**, & Sternberg, S. (2017). Zweitstimme.org. Ein strukturell-dynamisches Vorhersagemodell für Bundestagswahlen. *Politische Vierteljahresschrift*, 58(3): 418–441.

Working Papers and Articles Under Review

- **Neunhoeffer, M.** Learning from Small Data: Tabular Foundation Models for Political Science. Working paper.
- **Neunhoeffer, M.** Diversity Profiles for Party Systems. Under review.
- **Neunhoeffer, M.,** Schierholz, M., von Bartenwerffer, J., Beck, J., Henninger, F., Kestel, L., König, M., Seibold, H., Weber, W., & Kreuter, F. Learning by Solving: Building Government Data Literacy One Problem at a Time. Under review.
- **Neunhoeffer, M.,** Latner, J., & Drechsler, J. When Privacy Measures Fail: A Theoretical Critique of Empirical Disclosure Risk Assessment for Synthetic Data. Under review.

Work in Progress

- **Neunhoeffer, M.,** Sheldon, D., & Smith, A. A Bootstrap-Based General-Purpose Approach for Statistical Inference with Differential Privacy.
- **Neunhoeffer, M.,** Hod, S., & Drechsler, J. To Count or Not to Count: Practical DP Mean Estimation with Unknown Dataset Size.
- **Neunhoeffer, M.,** Sirvent, M., & Schierholz, M. What Information Do Job Embeddings Contain About Occupational Structure? Probing Against the German Occupational Classification.
- **Neunhoeffer, M.** A Common Benchmark to Evaluate Multiple Imputation Algorithms (MIBench).
- Sirvent, M. & **Neunhoeffer, M.** Uncertainty-Aware Embeddings for Occupational Similarity: Evidence from German Labor Market Data.

Other Publications

- **Neunhoeffer, M.** (2018). Book Review: Diana C. Mutz, In-Your-Face Politics: The Consequences of Uncivil Media. *Political Studies Review*, 16(1): NP76.

OPEN-SOURCE SOFTWARE

tabfound: Tabular Foundation Models in Pure R Torch | Author | github.com/mneunhoe/tabfound
Inference-only R implementations of open-weight tabular foundation models (TabPFN v2 to v3.5, Google TabFM, TabICL, Mitra) with no Python dependency.

RGAN: Generative Adversarial Nets in R | Author and maintainer | CRAN | github.com/mneunhoe/RGAN
Makes GANs accessible to social scientists in native R torch. 10,200+ CRAN downloads.

MIBench: Multiple Imputation Benchmark | Author | github.com/mneunhoe/MIBench
Standardized benchmark for multiple imputation algorithms across 11 scenarios and 9 methods.

partyscape | Author | github.com/mneunhoe/partyscape
Diversity profiles for party systems, accompanying the paper of the same name.

Private Post-GAN Boosting | Author | github.com/mneunhoe/post-gan-boosting
Replication code for the ICLR 2021 paper.

PUBLIC ENGAGEMENT AND TRANSLATIONAL WORK

zweitstimme.org, German Federal Election Forecast | Co-founder 2017–present
Public Bayesian forecasting platform combining polls and structural fundamentals. Forecast the 2017, 2021, and 2025 federal elections with calibrated uncertainty. Cov-

ered by The Washington Post, Süddeutsche Zeitung, Zeit Online, and Tagesspiegel.

ADA Bayern, Applied Data Analytics for Public Administration | Co-lead 2023–present
Data-literacy initiative co-led with colleagues in the Social Data Science and AI Lab at LMU Munich, in partnership with and funded by the Bavarian State Ministry for Digital Affairs. Co-designed tools with the Bavarian State Archives (deployed sampling application), the Bavarian Ministry of Education, and the Bavarian Forest National Park.

Op-eds: The Washington Post (Monkey Cage, 2017), Zeit Online (2018), Tagesspiegel (2017, 2019), Verfassungsblog (2020).

GOVERNMENT AND INSTITUTIONAL COLLABORATIONS

US Census Bureau 2021–2023
Differentially private methods for official statistics.

German Federal Statistical Office (Destatis) 2023–present
Privacy-preserving synthetic data for official statistics.

INVITED TALKS

Danish Statistical Society (DSTS) Two-Day Meeting on Clinical Data Science, Aalborg (2026); Italian Conference on Survey Methodology (ITACOSM), Bologna (2025); NBER Conference on Data Privacy Protection and the Conduct of Applied Research (2023, 2024); Workshop on Economics of Privacy and Data Labor, ICML (2020); Social Science Data Lab, Mannheim (2020); International Methods Colloquium (2019)

CONFERENCE PRESENTATIONS

Computer Science: ICLR (2021), Theory and Practice of Differential Privacy (2020, 2023, 2024, 2026), PODS (2024)

Statistics: Joint Statistical Meetings (2022, 2023, 2026), Privacy in Statistical Databases (2024, 2026), DAGStat (2019)

Social Sciences: PolMeth Europe (2021), EPSA (2017, 2019), APSA (2018), PolMeth (2018), PSA Methods (2019), ECPR (2017)

HONORS AND AWARDS

Publication of the Year Award 2018
CDSS and GESS, University of Mannheim, for “How Cross-Validation Can Go Wrong and What to Do About It.” EUR 1,000.

IPID4all Travel Grant 2018
DAAD, for the research stay at UC Berkeley. EUR 3,739.

TEACHING

University Courses

Diplomatische Akademie Wien: Machine Learning, Graduate (Spring 2026)

LMU Munich: Applied Data Analytics for the Public Sector, Graduate and Professional (Fall 2023–Fall 2026); Data and Privacy: Theoretical and Empirical Challenges, Graduate (Spring 2022); Statistics II for Sociologists, Undergraduate (Spring 2021)

UC Berkeley, School of Information: Privacy Engineering, Graduate (Summer 2020)

University of Mannheim: Tutorials, Multivariate Analyses, Graduate (Fall 2016–2020); Tutorials, Advanced Quantitative Methods, Graduate (Spring 2017–2020)

University of Applied Sciences Ludwigshafen: Applied Marketing Research, Graduate (Spring 2017)

Professional Training

Generative Adversarial Networks, Hertie School Data Science Summer School (2022); Introduction to Machine Learning, CIVICA Data Science Summer School (2021); Big Data and Social Science, GRADE Frankfurt (2019); Supervised and Unsupervised Machine Learning and Deep Learning, Deutsche Bundesbank (2019); Introduction to R, GQH Hessen (2018)

Supervision

Katharina Julia Brenner (M.A., LMU Munich, 2024): “Didaktische Ansätze für Stichprobentheorie im Archivwesen.” Grade: 1.0. Resulted in a blended-learning course now deployed at the Bavarian State Archives.

PROFESSIONAL SERVICE

Peer Review: Harvard Data Science Review, ICLR, NeurIPS, Theory and Practice of Differential Privacy, American Political Science Review, British Journal of Political Science, Journal of Politics, Political Analysis, Political Science Research and Methods, Party Politics, Public Opinion Quarterly, Politische Vierteljahresschrift, International Journal of Forecasting, Journal of Clinical and Translational Science

Conference Organization: Organizing Committee, PolMeth Europe 2021; Program Committee, IEEE SaTML 2027

TECHNICAL SKILLS

Software: R, Python, PyTorch, torch for R, TensorFlow, Stan, SQL, Git

Methods: Differential privacy, synthetic data, multiple imputation, tabular foundation models, Bayesian statistics, causal inference, double machine learning, conformal prediction, survey methodology, election forecasting

Languages: German (native), English (fluent), French (basic)